

Continued from last time

$(L \circ K): \mathbb{R} \rightarrow \mathbb{R}$ is linear Therefore there should be a single matrix representing $(L \circ K)$

$$\underset{\substack{1 \times 1 \\ \text{Cuz } \mathbb{R} \rightarrow \mathbb{R}}}{1 \times 1} \longrightarrow [1][x] = [x]$$

Can we obtain the matrix for $L \circ K$ from the matrices of L and K ? yes $[1 \ 0] \begin{bmatrix} 1 \\ 2 \end{bmatrix} = [1]$

K, L are linear

Thrm: if $\dim(\text{codomain of } K) = \dim(\text{domain of } L)$ then $L \circ K$
 1) exists 2) is linear 3) has a matrix representation computed using matrix multiplication

Ex:

$$\begin{array}{ccc} \text{"K"} & \text{"L"} & \\ \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix} & \begin{bmatrix} 7 & 9 \\ 8 & 0 \end{bmatrix} & \begin{array}{ccc} \mathbb{R}^3 & & \mathbb{R}^2 \\ \cong & K & \cong \\ A \rightarrow B & & C \rightarrow D \\ \text{over } \mathbb{R} & & \text{over } \mathbb{R} \end{array} \end{array}$$

is there a composition that makes sense?

$(K \circ L)$

$$\begin{array}{ccc} \mathbb{R}^2 & \xrightarrow{L} & \mathbb{R}^2 \\ \mathbb{R}^3 & \rightarrow & \mathbb{R}^2 \\ \uparrow & & \\ \text{X} & & \end{array}$$

$(L \circ K)$

$$\mathbb{R}^3 \xrightarrow{K} \mathbb{R}^2 \xrightarrow{L} \mathbb{R}^2$$

works!

$(L \circ K)$

$$\begin{bmatrix} 7 & 9 \\ 8 & 0 \end{bmatrix} \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix} = \begin{bmatrix} 7+18 & 21+36 & 35+54 \\ 8 & 24 & 40 \end{bmatrix} = \begin{bmatrix} 25 & 57 & 89 \\ 8 & 24 & 40 \end{bmatrix}$$